Sheet (2) Newton's law of Motion

1) A 3.00-kg object undergoes an acceleration given by $\bar{A} = 2.0\ i + 5.0\ j$ Find

(a) The resultant force acting on the object and

(b) The magnitude of the resultant force.

2) A 5.0-kg mass has the following forces on it:

$$F_1 = (4.0\ N)\ i - (3.0\ N)\ j$$

$$F_2 = (2.0\ N)\ i - (5.0\ N)\ j$$

Find the magnitude and direction of the acceleration of this mass.

3) A car with a mass of 850 kg is moving to the right with a constant speed of 1.44 m/s. What is the total force on the car? (b) What is the total force on the car if it is moving to the left?

4) A heavy freight train has a total mass of 16 000 metric tons. The locomotive exerts a pull of 670 000 N on this train. What is the acceleration? How long does it take to increase the speed from 0 to 50 km/h?

5) What average Force is required to bring a 1500 kg car to rest from a speed of 100 km/h with a constant acceleration within a distance of 55 m?

6) An electron of mass $9.11 * 10^{-31}$ has an initial speed of $3 * 10^5\ m/s$. it travels in a straight line and its speed increase to $7 * 10^5$ in 5 cm. Assuming constant acceleration, Calculate:

(a) The force exerted on the electron

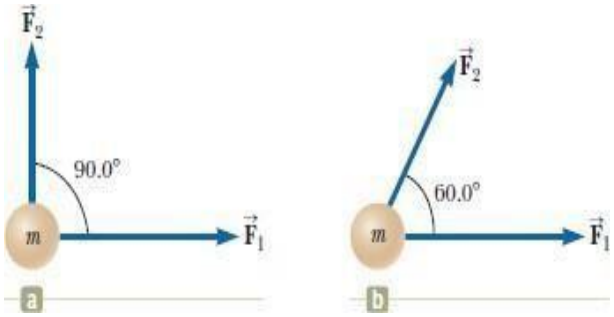
(b) Compare this force to the weight of the electron, which we neglected.

7) A force F is applied to an object of mass m_1 produces an acceleration of 3 m/s^2. The same force is applied to a second object of mass m_2 produces an acceleration of 1 m/s^2.

(a) What is the value of the ratio m_1/m_2?

(b) If m_1 and m_1 are combined into one object, find its acceleration under the action of the force F.

8) Two forces F_1 and F_2 act on a 5.0-kg object. Taking $F_1 = 20.0\ N$ and $F_2 = 15.0\ N$, find the accelerations of the object for the forces shown in the figure.



9) A man weighs 750 N on the Earth. The surface gravities of Mars and Jupiter are 0.38g and 2.53g, respectively. What is the mass of man on each of these planets? What is his weight on each planet?

10) Three rescuers are pulling horizontally on a safety net to keep it taut. One pulls northward with a force of 270 N; the second pulls in a direction 30° south of west with a force of 240 N. In which direction and with what force must the third pull to keep the net stationary?

11) An ocean current applies a force of 2500 N to a 1400-kg sailboat in a direction 15°east of north. The wind applies a force of 3200 N in a direction 30° east of north. What are the magnitude and direction of the resulting acceleration?